PART 1

9

EXPERIMENT 9

Study of Refraction of Light Through a Glass Prism

Aim

To study the refraction of light through a glass prism by tracing the path of rays of light passing through the prism and measuring the angle of deviation for different angles of incidence.

Requirements

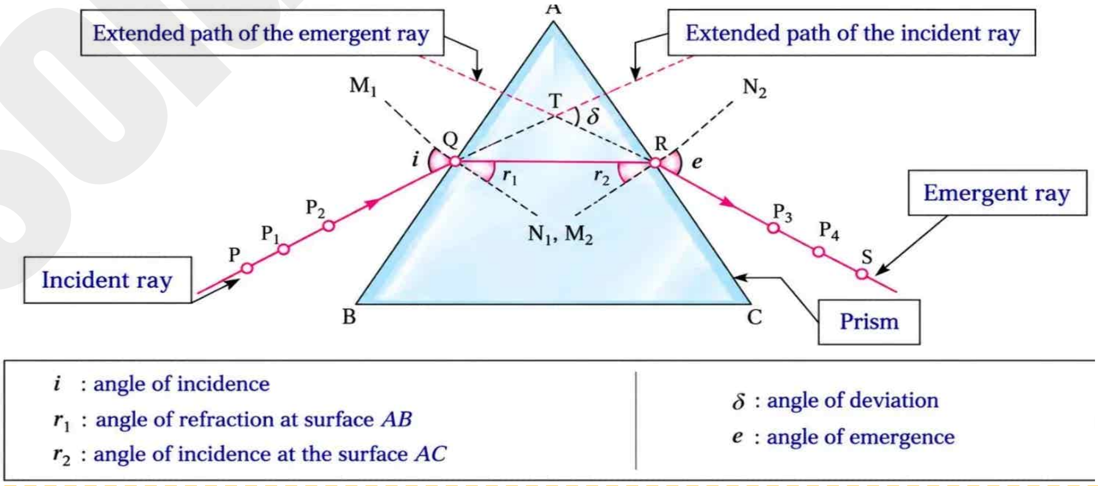
A glass prism, a drawing board, a white drawing paper, paper pins and drawing pins, and a protractor.

Procedure

1. Fix the drawing paper onto the drawing board using drawing pins.
2. Place the glass prism on the paper with its triangular base touching the paper, and trace its boundary with a sharp pencil.
3. Remove the prism and draw a normal M_1N_1 at point Q. Draw a ray PQ that makes an angle of incidence (i) of 30° with the normal M_1N_1.
4. Fix pins P_1 and P_2 on ray PQ so that the pins stand vertical and are well separated.
5. Place the prism back in its original position. Looking through the prism from the opposite side, view the images of the bases of pins P_1 and P_2, and fix two more pins P_3 and P_4 such that their bases line up exactly with these images.
6. Remove pins P_3 and P_4 and mark their positions with small circles. Draw a line through points P_3 and P_4, and let it meet the surface AC of the prism at point R. Draw ray RS by taking a point S on the line joining P_3 and P_4. Remove pins P_1 and P_2 along with the prism. Join QR and draw a normal M_2N_2 at point R.
7. Extend ray PQ (the incident ray) in the same direction, and ray RS (the emergent ray) in the opposite direction, so that they meet each other at point T.
8. Measure the angle of incidence (i), the angle of refraction (r_1), the angle of deviation (δ), and the angle of emergence (e), and record these in the observation table.
9. Repeat the same procedure for angles of incidence of 45° and 60°.

Diagram

(Label the various components in the given diagram - Refraction of light through a glass prism)



Observation

The path of the ray for angle of incidence of 30°/45°/60° is PQRS.

Observation Table

Angle of Incidence (i)	Angle of Refraction (r_1)	Angle of Deviation (δ)	Angle of Emergence (e)
30°	20°	43°	73°
45°	30°	37°	60°
60°	38°	40°	40°

Inferences

1. The angle of incidence (i) > the angle of refraction (r_1). This means that when a ray of light travels from air to glass, it bends towards the normal.
2. The angle of emergence (e) > the angle r_2. This means that when a ray of light travels from glass to air, it bends away from the normal.
3. In general, the angle of deviation (δ) depends on the angle of incidence (i). If the angle of incidence is gradually increased, starting from (say) 30°, initially δ decreases as i is increased, becomes minimum for a certain value of i, and then increases as i is further increased.

Precaution

Pins should be fixed such that they are vertical and well separated.

Multiple Choice Questions

1. The change in the direction of propagation of light when it passes obliquely from one transparent medium to another is called ....

(A) dispersion (B) scattering (C) refraction (D) reflection

Answer: (C) refraction

2. A ray of light gets deviated when it passes obliquely from one medium to another medium because ....

(A) the colour of light changes (B) the frequency of light changes
(C) the speed of light changes (D) the intensity of light changes

Answer: (C) the speed of light changes

3. .... of light is responsible for twinkling of stars.

(A) Reflection (B) Internal reflection (C) Dispersion (D) Refraction

Answer: (D) Refraction

4. .... light is deviated the maximum in the spectrum of white light obtained with a glass prism.

(A) Red (B) Yellow (C) Violet (D) Blue

Answer: (C) Violet

5. .... light is deviated the least in the spectrum of white light obtained with a glass prism.

(A) Red (B) Yellow (C) Violet (D) Blue

Answer: (A) Red

Remember

Even when the incident rays are directed away from the base of the prism, the emergent rays bend towards the base of the prism, as the prism is triangular.

Date: ______________ Teacher's Signature: ______________